Unit IV-The Network Layer

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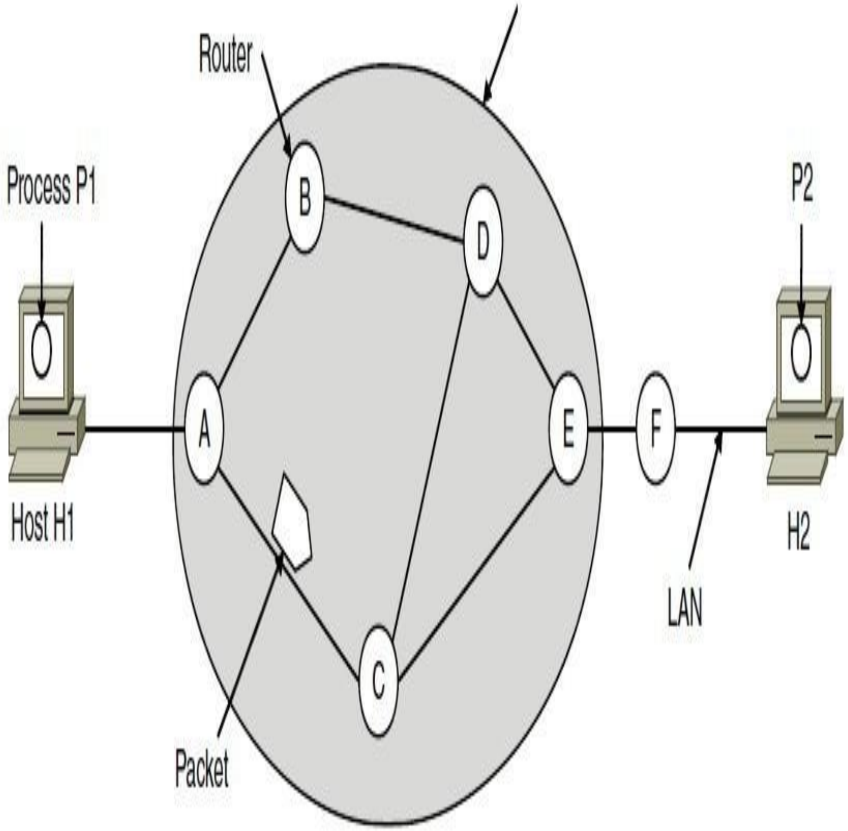
Network Layer Design Issues

- Store-and-forward packet switching
- Services provided to transport layer
- Implementation of connectionless service
- Implementation of connection-oriented service
- Comparison of virtual-circuit and datagram networks

Store - and - forward packet



The network layer operates in an environment that uses store-and-forward packet switching.

Store-and-forward packet switching is a technique in which data packets are temporarily stored at each intermediate node before being forwarded to the next node.

Each intermediate node verifies whether the packet is error-free prior to transmission, thereby ensuring the integrity of the data packets.

Advantages and Disadvantages

Advantages

- Ensures high-quality data packet transmission.
- Erroneous packets are detected and discarded at each router.
- As a result, faulty or invalid packets are effectively eliminated from the network.

Disadvantages

- Error-free packet transmission is ensured in store-and-forward switching.
- However, this reliability is achieved at the cost of reduced transmission speed.
- Switch latency occurs because each node must wait for the complete packet to arrive before forwarding.
- Additional delay is introduced due to the computation of the CRC (Cyclic Redundancy Check).
- While the latency at a single router may be minimal, the cumulative delay across multiple routers becomes significant.
- Therefore, store-and-forward packet switching is not suitable for time-critical or real-time online applications.

Services Provided to the Transport Layer

1.Services independent of router technology.

2.Transport layer shielded from number, type,topology of routers.

3.Network addresses available to transport layer are uniform even across LANs and WANs

A Connectionless service

- A Connectionless service is a data communication between two nodes where the sender sends data without ensuring whether the receiver is available to receive the data.
- Here, each data packet has the destination address and is routed independently irrespective of the other packets. Thus the data packets may follow different paths to reach the destination. There's no need to setup connection before sending a message and it after the message has been sent.
- The data packets in a connectionless service are usually called datagrams. Network is called datagram network.
- Protocols for connectionless services are −
    - ➢Internet Protocol (IP)
    - ➢User Datagram Protocol (UDP)
    - ➢Internet Control Message Protocol (ICMP)

Advantages & Disadvantages

Advantages:

- Provides very fast data transmission.
- Supports multicast and broadcast operations, enabling the transfer of similar data to multiple recipients in a single transmission.
- Errors can be minimized by implementing error-correcting mechanisms within the application protocol.
- Simple and low-overhead service, making it easy to use.
- The network layer host software is relatively simpler.
- No authentication process is required.
- Suitable for applications that do not require sequential delivery of packets, such as packet voice and real-time communications.

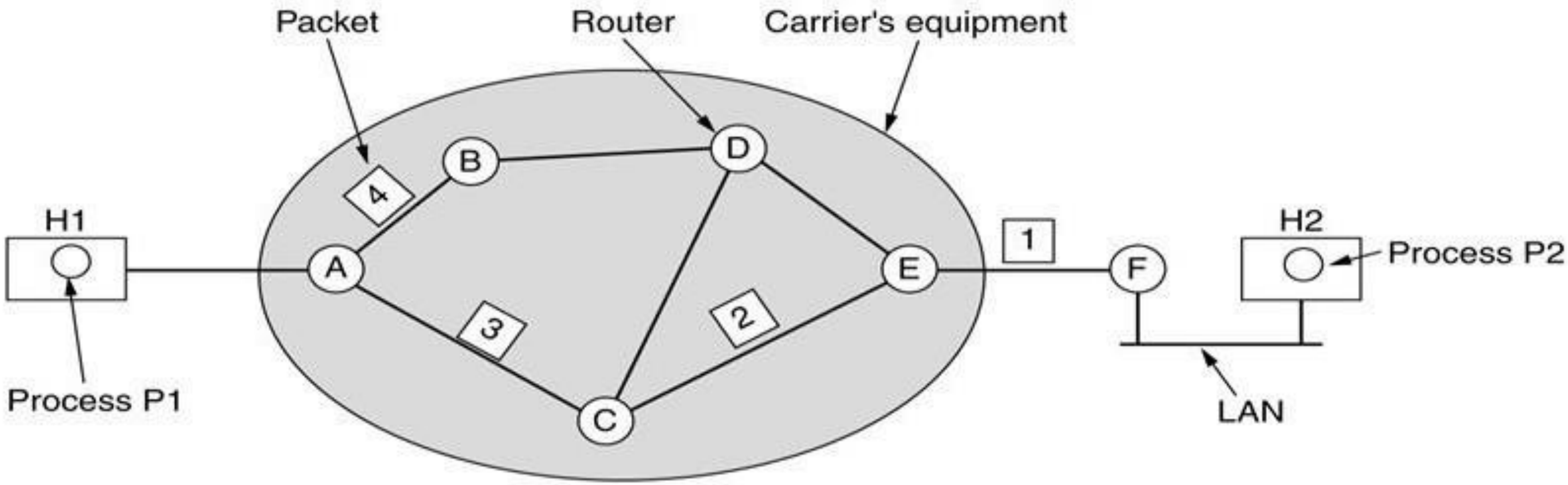
Disadvantages:

- Less reliable compared to connection-oriented services.
- Does not guarantee error-free, loss-free, or in-sequence packet delivery.
- Packets may experience loss, duplication, misdelivery, or arrive out of order.
- More prone to network congestion under heavy traffic conditions.

Routing Principle

- Packet switching networks use datagram network.
- Datagrams are data packets which contain adequate header information so that they can be individually routed by all intermediate switching devices to the destination.
- These networks are called datagram networks since communication occurs via datagrams.
- In datagram networks, each data packet is routed independently from the source to the destination even if they belong to the same message.
- No prior resource or channel allocation is done for the individual packets. As the datagrams are treated as independent units, no dedicated path is fixed for data transfer. Each datagram is routed by the intermediate routers using dynamically changing routing tables.
- So two successive packets from the source may follow completely separate routes to reach destination. Resources are allocated on demand on a First- Come First-Serve (FCFS) basis. When a packet arrives at a router, the packet must wait if there are other packets being processed, irrespective of its source or destination.

Implementation of Connectionless Service



Dest.	A's table initially (Line)	A's table later (Line)	C's table	E's table
A	–	–	A	C
B	B	B	A	D
C	C	C	–	C
D	B	B	D	D
E	C	B	E	–
F	C	B	E	F

Routing within a diagram subnet.

Implementation of Connection-Oriented Service

- A connection-oriented service is one that establishes a dedicated connection between the communicating entities before data communication commences. It is modeled after the telephone system.
- To use a connection-oriented service, the user first establishes a connection, uses it and then releases it. In connection-oriented services, the data streams/packets are delivered to the receiver in the same order in which they have been sent by the sender.
- Connection-oriented services may be done in either of the following ways
  Circuit- switched connection: In circuit switching, a dedicated physical path or a circuit is established between the communicating nodes and then data stream is transferred.
  Virtual circuit-switched connection: the data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver.
  A virtual path is established and Network is called virtual -circuit network

Advantages & Disadvantages of Connection-Oriented Services

- This is mostly a reliable connection.
- Congestions are less frequent.
- Sequencing of data packets is guaranteed.
- No duplicate data packets
- Suitable for long connection.

Disadvantages

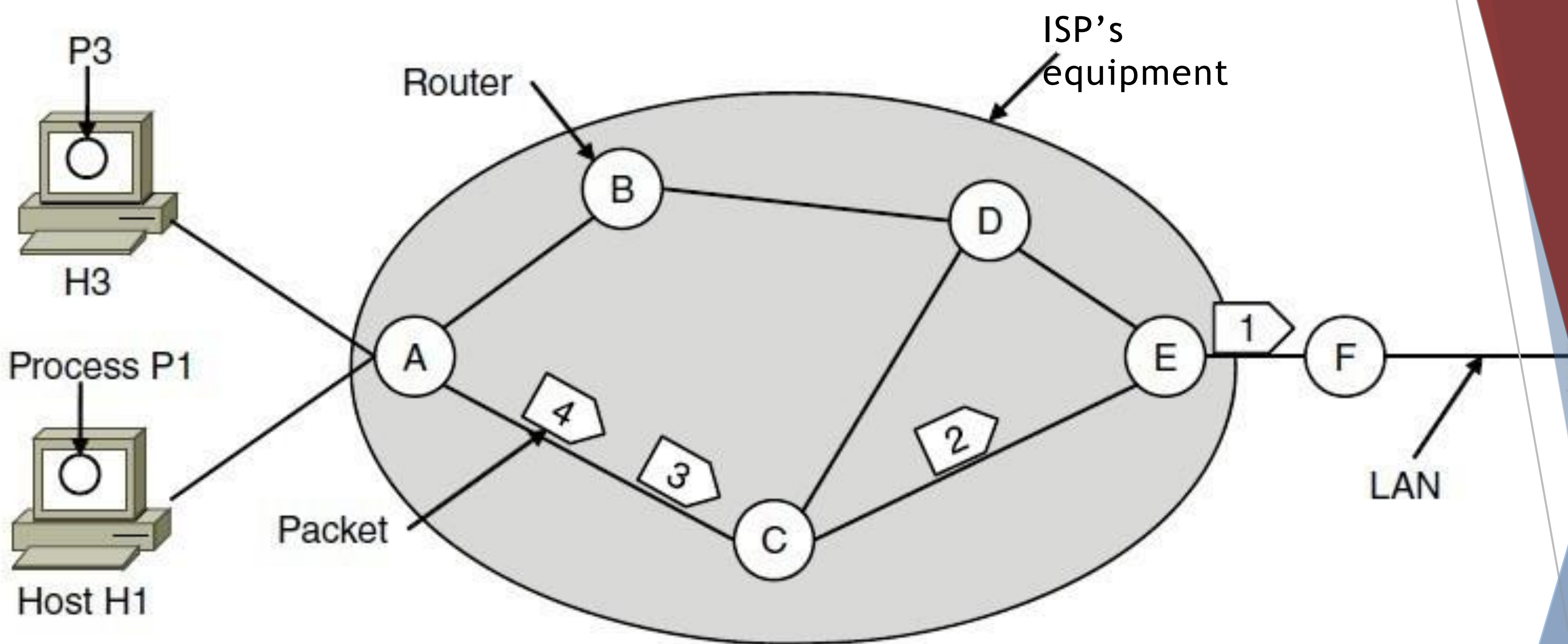
- Resource allocation is needed before communication. This often leads to under-utilized network resources.
- The lesser speed of connection due to the time is taken for establishing the connection.
- In the case of router failures or network congestions, there are no alternative ways to continue communication.

Phases of Virtual - Circuit Transmission

In a virtual circuit network, data transmission occurs in three distinct phases:

1. **Connection Establishment:**
    - A virtual circuit (route) is established between the source and destination through a series of switches.
    - The source and destination use global addresses, which help switches create routing table entries for the connection.
2. **Data Transfer:**
    - Once the virtual circuit is set up, all packets follow the predefined route established during the setup phase.
    - Routing tables ensure that packets are delivered in sequence and along the same path.
3. **Connection Release:**
    - After the data transfer is completed, the source sends a teardown request.
    - The destination replies with a teardown confirmation.
    - All switches involved clear their routing table entries, releasing the resources allocated for the virtual circuit.

Implementation of Connection-Oriented Service



A's table

In		Out	
H1	1	C	1
H3	1	C	2

C's Table

A	1	E	1
A	2	E	2

E's Table

C	1	F	1
C	2	F	2

Routing within a virtual-circuit network

Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Comparison of datagram and virtual-circuit networks

Circuit Switching Vs Packet Switching

Circuit Switching	Packet Switching
Physical path between source and destination	No physical path
All packets use same path	Packets travel independently
Reserve the entire bandwidth in advance	Does not reserve
Bandwidth Wastage	No Bandwidth wastage
No store and forward transmission	Supports store and forward transmission

Routing Algorithms

- Routing is the process by which the network layer determines the path that packets should take from the source to the destination.
- The network layer decides the **output line** through which an incoming packet should be transmitted.
- To ensure successful delivery, the **best route** or **least-cost path** must be selected for packet transmission.
- The **routing protocol** performs this function — it uses routing algorithms to determine the most efficient path from source to destination.
- Regardless of whether the network layer uses **datagram service** or **virtual circuit service**, its main responsibility remains to find the optimal route for packet delivery.
- **Key Points:**
- In a **datagram network**, the route may vary for each packet, as each packet is treated independently.
- In a **virtual circuit network**, the route is determined during the setup phase and remains fixed throughout the session — this is known as **session routing**

Routing Algorithms

- **Functions of Routing:**

•**Packet Forwarding:** Handles each incoming packet and directs it toward the appropriate next hop or destination.

•**Routing Table Management:** Creates, fills, and updates routing tables to ensure accurate and up-to-date path information.

Goals of Routing:

1. **Simplicity:** The routing process should be simple and introduce minimal overhead.
2. **Correctness:** Routes must be accurate and reliable.
3. **Robustness:** The system should remain effective even when conditions or network topology change.
4. **Stability:** Routing should remain consistent and steady under varying network situations.
5. **Fairness:** Every node should get a fair opportunity to transmit packets.
6. **Optimality:** Minimize packet delay and maximize throughput.

Classification of a Routing algorithm

Adaptive Routing algorithm

- An adaptive routing algorithm is also known as dynamic routing algorithm.
- This algorithm makes the routing decisions based on the topology and network traffic.
- The main parameters related to this algorithm are hop count, distance and estimated transit time.

Non-adaptive Routing algorithm

- Non Adaptive routing algorithm is also known as a static routing algorithm.
- When booting up the network, the routing information stores to the routers.
- Non Adaptive routing algorithms do not take the routing decision based on the network topology or network traffic.

Types of adaptive routing techniques

.

- **Centralized Algorithm:** Determines the least-cost path between source and destination using complete (global) network information, hence also called the *global routing algorithm*.

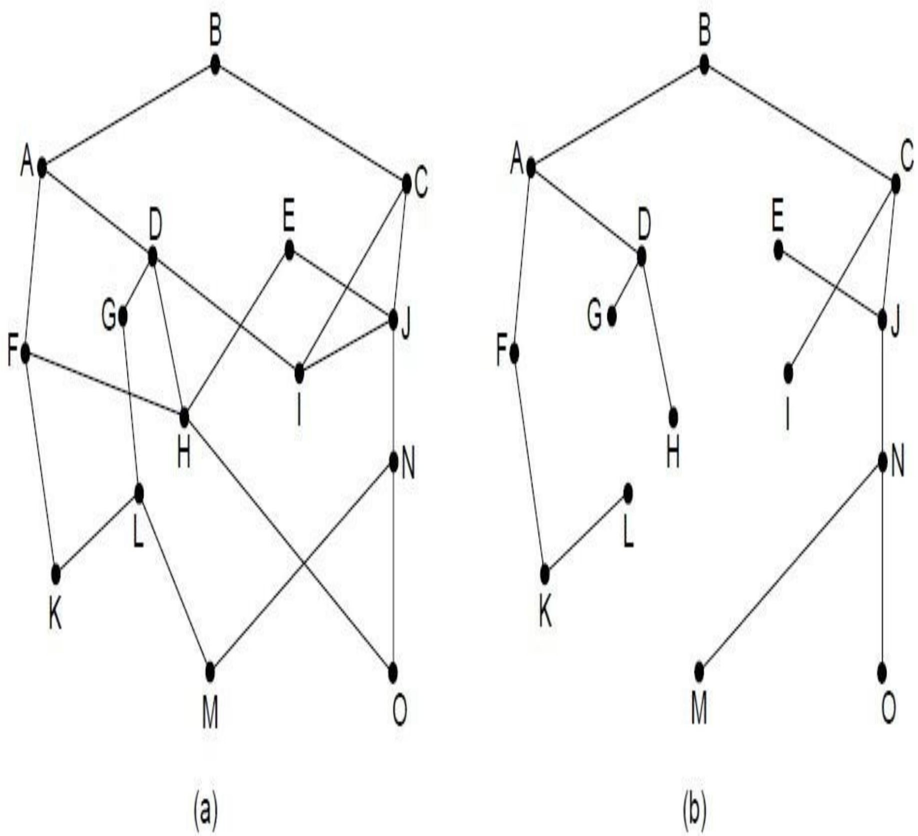
•**Isolated Algorithm:** Makes routing decisions using only local information, without exchanging data with other nodes.

•**Distributed Algorithm:** A decentralized method where each node collaborates and iteratively computes the least-cost path between source and destination.

Routing Algorithms

- Optimality principle
- Shortest path algorithm
- Flooding
- Distance vector routing
- Link state routing
- Routing in ad hoc networks- Hierarchical Routing
- Broadcast routing
- Multicast routing

The Optimality Principle



(a) A network. (b) A sink tree for router *B.*

It is a principle about optimal routes without regard to network topology or traffic

Principle :If router J is the optimal path from router I to router K then optimal path from J to K also falls along same route.

Sink Tree

- Tree containing optimal routes from all sources to destination
- Root is destination
- Contains No loops
- Packet delivered in a finite & bouned number of hops
- Not unique

Shortest Path Algorithm

- Static routing algorithms
- The subnet is represented as a **graph**, where each **node** denotes a router and each **arc** represents a communication link or line.

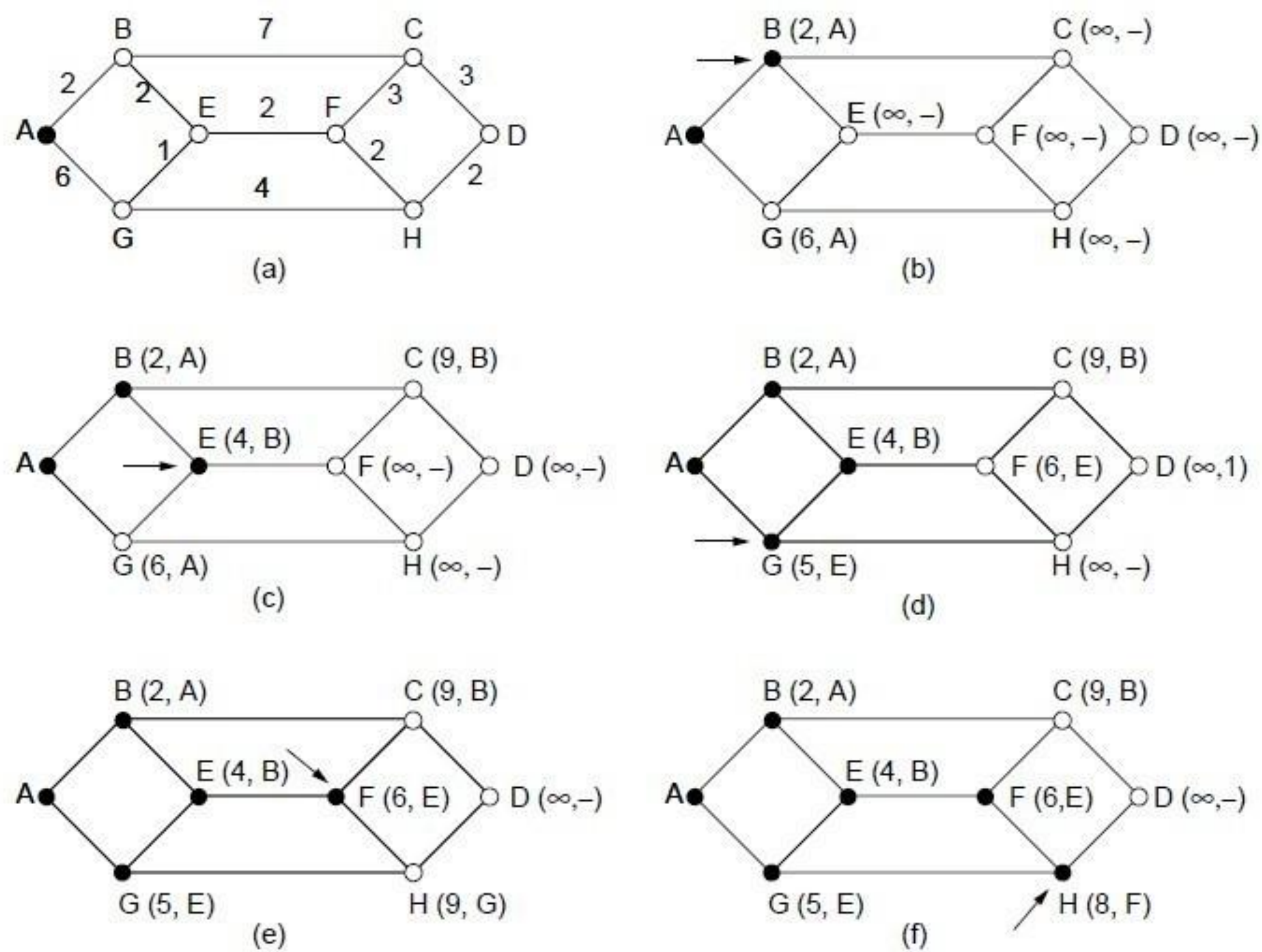
Some of the well-known shortest path algorithms are:

- **Bellman-Ford Algorithm**
- **Dijkstra’s Algorithm**
- **Floyd-Warshall Algorithm**

•To determine the route between any two nodes, the algorithm identifies the **shortest path** connecting them.

•Initially, since no paths are known, all nodes are labeled with **infinity** until paths are computed.

Shortest Path Algorithm



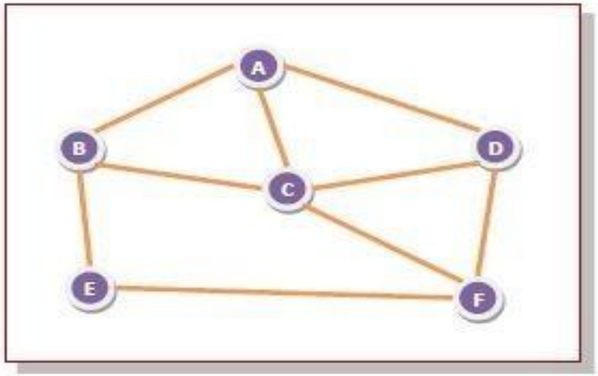
The first five steps used in computing the shortest path from *A to D.*

The arrows indicate the working node

Non-Adaptive (Static) Routing - Flooding

Technique:
In this method, when a data packet arrives at a router, it is forwarded through all outgoing links **except** the one it arrived on.

For a network with six interconnect



•When a packet reaches **Router A**,
B, C, and D.
•**Router B** sends the packet to **C a**
•**Router C** forwards it to **B, D, and**
•**Router D** sends it to **C and F**.
•**Router E** forwards the packet to **F**.
•**Router F** sends it to **C and E**.

This approach ensures delivery but may lead to redundant transmissions and congestion due to multiple copies of the same packet circulating in the network.

Advantages of Flooding

1. **Simplicity of Implementation:** It is very simple to set up and implement since each router only needs to know its immediate neighbors.
2. **High Reliability:** The flooding technique is highly robust; even if several routers fail, data packets can still reach the destination through alternate paths.
3. **Comprehensive Coverage:** All nodes that are directly or indirectly connected are visited, ensuring that no node is left out—making it ideal for broadcasting.
4. **Optimal Path Selection:** Flooding ensures that the data packets always follow the shortest available path to the destination.

Disadvantages of Flooding:

1. Flooding can generate an excessive number of duplicate packets, leading to network congestion.
2. It is inefficient for communication with a single destination since data packets are delivered to all nodes, regardless of the target node.
3. The network may become overloaded with unnecessary and redundant packets.
4. Due to its inefficiency, this technique is not suitable for most practical applications.

Applications:

Flooding is mainly used in **military networks**, **distributed database systems**, and **wireless networks**, where reliability and complete data delivery are crucial.

Solutions to Control Duplicate Packets in Flooding:

Hop Counter:
Each packet contains a hop counter in its header, which is decremented at every hop along the path. When the counter reaches zero, the packet is discarded, preventing indefinite circulation in the network.

Sequence Numbering:
The source router assigns a unique sequence number to every packet it receives from hosts. Each router maintains a list of sequence numbers already processed from each source. If a packet with the same sequence number is received again, it is discarded to avoid duplication.

Selective Flooding:
Instead of sending packets on every outgoing line, routers forward packets only through links that approximately lead toward the destination. This reduces unnecessary traffic and optimizes network efficiency.

Distance Vector Routing

- More complex than flooding but far more efficient.
- Aims to find the **shortest path** for the current network topology using algorithms like the **Bellman-Ford Algorithm**.
- Each router maintains a **routing table (or vector)** containing the best known distance and the corresponding outgoing line to each destination.
- Every router in the network builds and maintains its own routing table to determine the optimal path through the network.
- Routers periodically **exchange and update** their tables based on information received from neighboring routers.
- **Delay** is measured using **ECHO packets**, which help assess link performance.
- **Routing metrics** may include parameters such as **delay, cost, time, hop count**, and **number of packets**.

Distance Vector Routing – Working Principle

•Each router periodically **sends its distance vector** to all its neighboring routers through a routing packet.

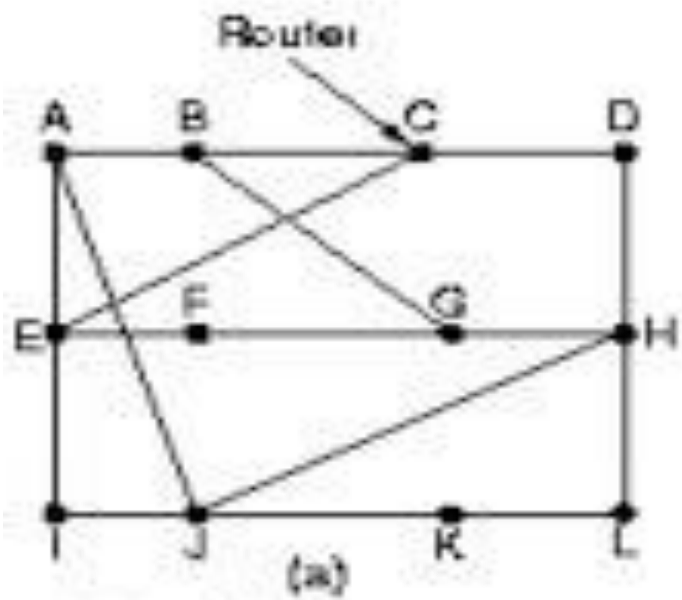
•Upon receiving updates, a router **stores the most recent distance vectors** from each of its neighbors.

•A router **recomputes its own distance vector** in the following cases:

•When it receives a new distance vector from a neighbor that contains **changed or updated information**.

•When it **detects a link failure** or finds that a connection to a neighbor has gone down.

Distance vector example



(a) A subnet.

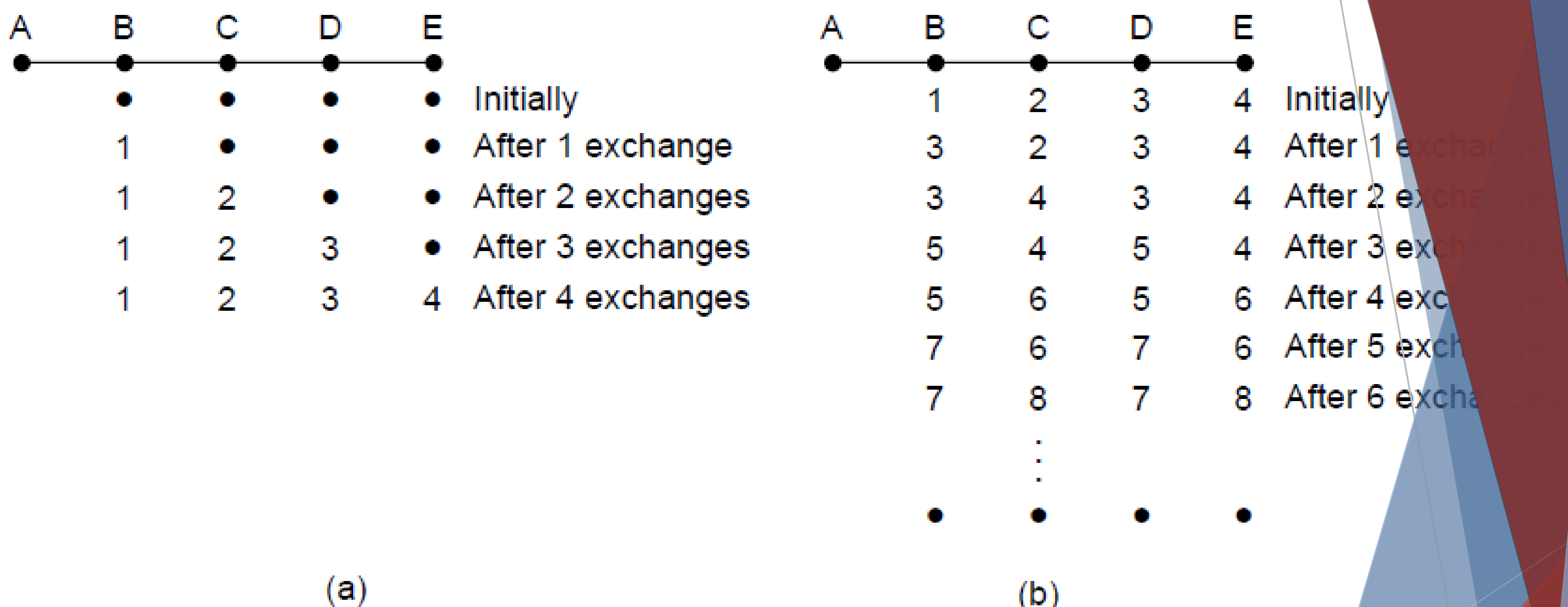
(b) Input from *A*, *I*, *H*, *K*, and the new routing table for *J*.

To	A	I	H	K	delay from J	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	–
K	24	22	22	0	6	K
L	29	33	9	9	15	K
	JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6	New routing table for J	

Vectors received from

(b)

The Count-to-Infinity Problem



The count-to-infinity problem

Advantages of Distance Vector routing –

It is simpler to configure and maintain than link state routing.

Disadvantages of Distance Vector routing –

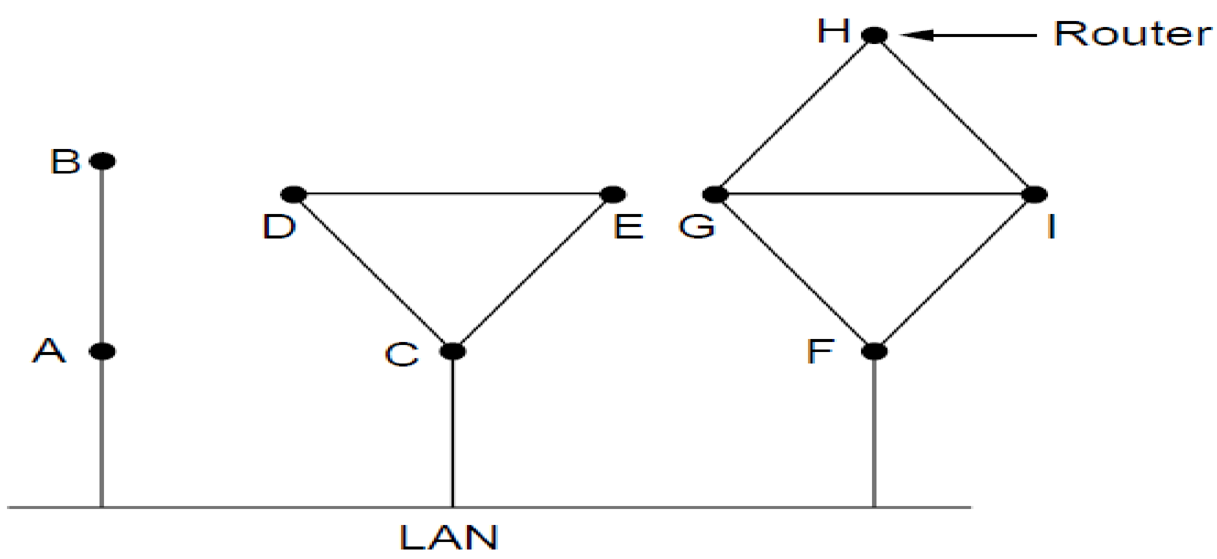
- It is slower to converge than link state.
- It is at risk from the count-to-infinity
- problem.
- Works in theory,has serous drawback in practice- Very slow
- It reacts fast to bad news

Link State Routing

1. Discover neighbors, learn network addresses.- **HELLO packet**
2. Measure the delay/cost metric to each neighbor.--- special **ECHO packet**
3. Construct packet all it has just learned.---**Link State packet**
4. **Distribute** LS packets --to receive packets from other routers.
5. Compute **new routes**---- shortest path to every other router.

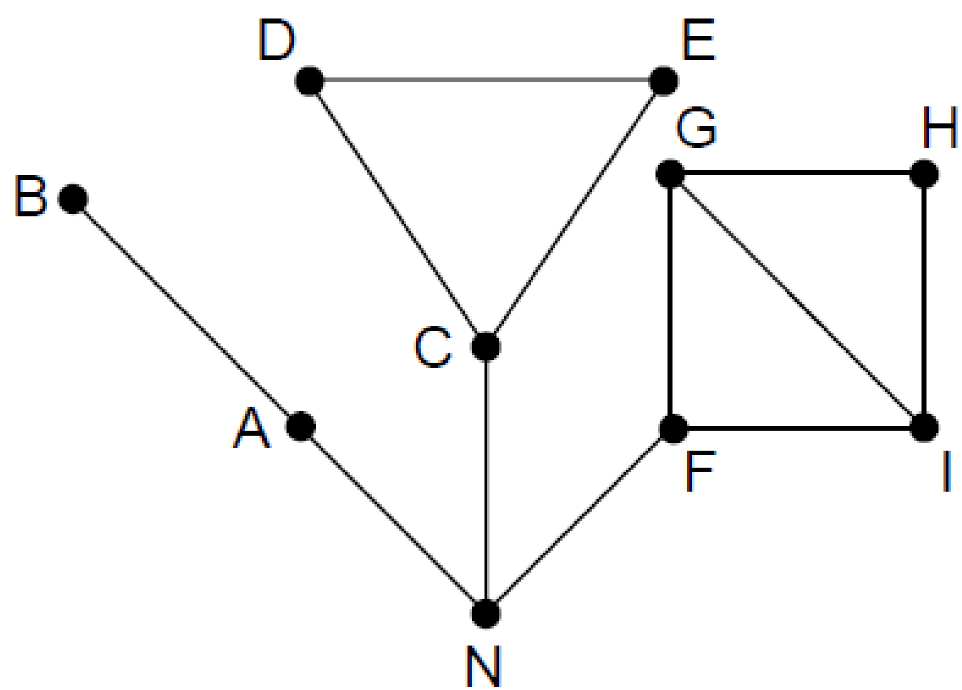
1. Learning about the Neighbors (1)

- Send HELLO packet
- Address should be specified properly



Nine routers and a broadcast LAN.

Learning about the Neighbors (2)



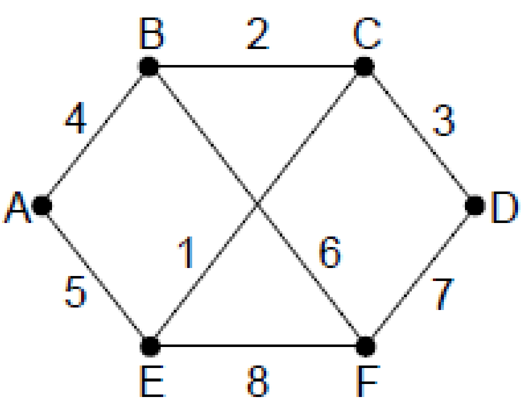
A graph model of previous slide.

2. Setting link costs

Finding shortest path

- Distance configured by operator
- Echo use to find the delay
- Round trip time and devide by 2 tocaluculate total delay

3. Building Link State Packets



(a)

Link State Packets

A		B		C		D		E		F	
Seq.		Seq.		Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	[illegible]
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1			F	8	E	8

(b)

(a) A network. (b) The link state packets for this network.

4. Distributing the Link State Packets

- Router must get packet quickly and reliably
- Routers using different version of topology->computr can inconsistencies
- Flooding
- Check each packet seq number and pair (source and seq)
- Check new or old packet
- If new forward to all expect arrived If dublicate discard

Problems faced during distribution of packets

- **1.Sequence number confusion** – If a router crashes and loses track of its last sequence number, it may send outdated LSPs that cause inconsistencies in the network.
  2.Corruption of LSPs – Errors during transmission can alter the sequence number or data, leading to incorrect topology information.
- **3. Old LSPs remaining in the network** – Without proper aging (time-to-live), outdated LSPs may persist and create routing errors.

solutions to the problems

1.Use of 32-bit sequence numbers – Provides a large range to prevent confusion or overlap when routers restart or send frequent updates.
2.Inclusion of an Age field – Each LSP has an age timer that decreases over time; when it reaches zero, the LSP is discarded to remove outdated information.

4. Distributing the Link State Packets

			Send flags			ACK flags			
Source	Seq.	Age	A	C	F	A	C	F	Data
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

The packet buffer for router *B* in previous slide

5. Computing the new routes

- Dijkstra algorithm – shortest path algorithm

Disadvantage

- More memory and computation

Hierarchical Routing

Table grows with respect to network growth

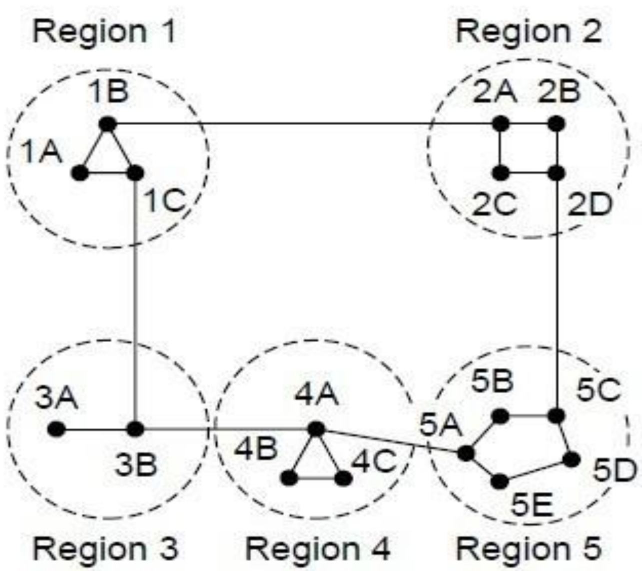
- Memory consumption
- CPU time to scan tables
- More bandwidth to send status report
- **Solution**—hierarchical –telephone n/w
- ***Regions***

Group regions into ***clusters***

Clusters into ***zones***

Zones into ***groups*** so on

Hierarchical Routing



(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

Hierarchical routing.

Hierarchical Routing Example

For example, the optimal path from router **1A to 5C** is through **region 2**. However, in hierarchical routing, all traffic destined for **region 5** is routed via **region 3**, as it provides a better overall route for most destinations within region 5.

Consider a subnet consisting of **720 routers**.

- If **no hierarchy** is implemented, each router must maintain **720 entries** in its routing table.
-

When the subnet is **divided into 24 regions**, each containing **30 routers**, then:

- Each router stores **30 local entries** (for routers within its region) and
- **23 remote entries** (for other regions),resulting in a total of **53 routing table entries**.

•**Further optimization:**

If the same **720-router subnet** is divided into **8 clusters**, each containing **9 regions**, and each region has **10 routers**, then:

- Each router maintains **10 local entries + 8 regional entries + 7 cluster entries**,
  giving a total of **25 entries** per routing table.

Broadcast Routing

- Host Sending informationto all other
- Host
- **Types of broadcasting**
- **1. distinct packet to each dest-** waste of BW
- **2. Flooding**-well suited

  but prob-too many duplicate Ps

3.Multi destination routing – P. has a **list of**
- **dest/bit map**. Router creates **new copy of P.**

  adds address of dest
-

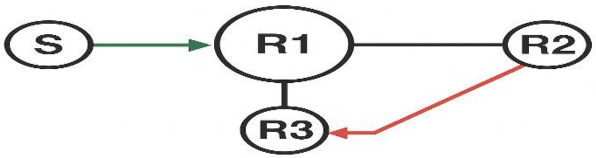
4. Spanning tree/sink tree—is subset of subnet include all routers without any loops

5. **Reverse path forwarding-** If a router receives a packet through its preferred (shortest) path back to the source, it forwards the packet to other interfaces; otherwise, it discards it. In this way, the packet itself follows the best route through the network.

Example

S = Source router

R1, R2, R3 = Other routers



Step-by-Step Explanation

S sends a broadcast packet to **R1**.

R1 receives the packet and forwards it to **R2** and **R3**.

R2 gets the packet from **R1** and checks:

Is **R1** on my shortest path back to **S**?

Yes ✅ → So, R2 **forwards** the packet to its other neighbors (if any).

R3 also gets the packet from **R1**, checks the same condition:

Is **R1** on my shortest path back to **S**?

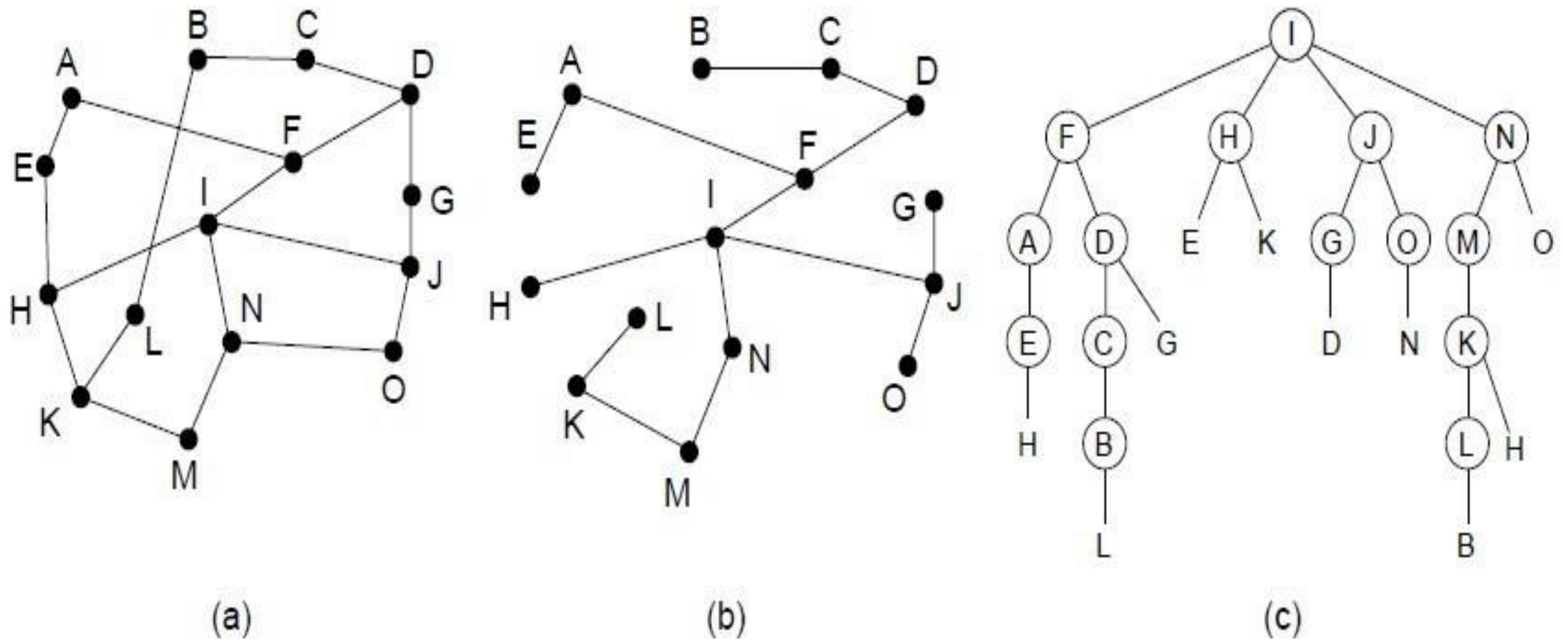
Yes ✅ → R3 **forwards** it further.

If **R2** later receives another copy of the same packet from **R3**, it checks:

Did it arrive on my shortest path to **S**?

No ❌ → **Discard the packet**.

Broadcast Routing



Reverse path forwarding. (a) A network. (b) A sink tree.

(c) The tree built by reverse path forwarding.

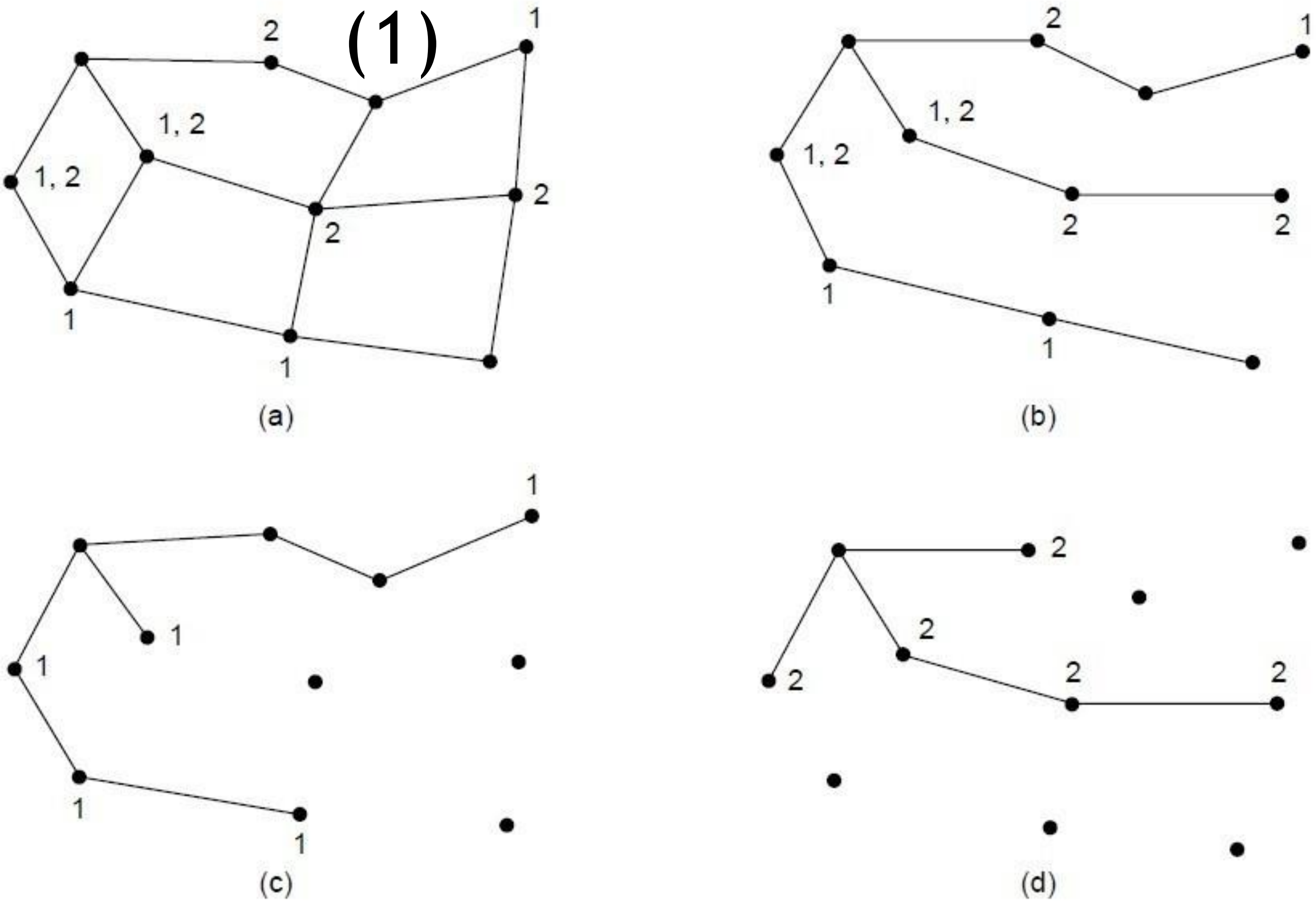
Multicast Routing

- Sending to Large Groups(1000 m/c)
  -Broadcasting is inefficient in a million node
- n/w Group- ***multicasting***
- Group management-**create,destroy**
- Processes can **join & leave groups**
- When P. arrives-Router examine spanning tree & **prune it**
- **Prune**---remove all line that dont go to goup members

Link state routing-spanning tree is pruned **distance vector routing**- reverse path forwarding **Disadv:** not efficient in very large n/w

For n groups containing m members- each group needs m spanning trees and **total mn trees**

Multicast Routing (1)



(a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.